Why We Started This Program

We started this program on Teachers' Day as a small way to express gratitude to all the teachers and mentors who have guided us, shared their knowledge, and helped us grow.

Instead of only wishing "Happy Teachers' Day," we wanted to do something meaningful. We believe the best way to respect the knowledge we have received is to share it with others.

This free Machine Learning course is started with the intention of helping students who genuinely want to learn, explore, and build something in the field of AI/ML, especially those who may not have access to proper guidance.

There is no commercial motive behind this initiative. Our goal is to provide practical and conceptual learning, continuous support, doubt-clearing, mentorship, and career guidance so students can grow with confidence.

This program is our small effort to give back, share knowledge, and begin a meaningful learning journey together.

What You Will Get in This Course

After completing this course, you will receive access to the following benefits:

- Access to our professional network and contacts.
- Practical projects with submission and evaluation.
- Regular updates about hackathons and competitions.
- Guidance and support for participating in hackathons.
- Help in building industry-level projects.
- Career guidance and mentorship.
- Interview preparation support.
- Support with referrals wherever possible.

Most Important

After this course, the top 10 performers will get an internship opportunity.

Once you connect with us, you will receive lifetime support from our side, not only during the course but even after completing it.

Our Target

Our goal is to make you industry-ready as quickly as possible. We will cover the mathematics of each algorithm in detail.

Our main focus is to help you start building machine learning projects quickly, participate in competitions, and gain practical industry-level skills.

Course Structure

The course duration will be 6 months and will cover the following topics:

- Python programming, including NumPy, Pandas, and Matplotlib.
- Core machine learning concepts to help you build and apply ML models effectively.
- Detailed mathematics behind machine learning algorithms.
- Introduction to deep learning.
- Placement preparation.
- Classroom hackathon.
- Internship opportunity for top performers.

Road Map to Become a Data Scientist / ML Engineer

Python for Data Science

↓

Data Processing

↓

Machine Learning

↓

Deep Learning

↓

SQL

Machine Learning Syllabus

Module 1: Introduction to Machine Learning

- Applications of Machine Learning
- Difference between Traditional Coding and Machine Learning
- Supervised vs Unsupervised vs Reinforcement Learning
- Types of Machine Learning: Regression and Classification

Module 2: Simple Linear Regression

- Simple Linear Regression
- Machine Learning Workflow

Module 3: Python Basics

- Python
- NumPy, Pandas
- Train-Test Split
- Overfitting and Underfitting Basic Idea

Module 4: Model Training, Validation, and Tuning

- Training Machine Learning Model: Linear Regression
- Train-Test Split
- Hyperparameter Tuning
- Regression Metrics

Module 5: Data Preprocessing and EDA

- Task 1
- Data Preprocessing
- Data Visualization
- Exploratory Data Analysis

Module 6: Feature Engineering

- Feature Engineering
- Feature Scaling
- Transformers
- Outliers
- Missing Values Handling
- Encoding
- Duplicate Value Handling
- Binning

- Date/Time Feature Extraction
- ColumnTransformer
- Task 2

Module 7: Mathematics and Optimization

- Integration and Differentiation
- Loss Function
- Gradient Descent and its Types
- Learning Rate
- Task 3

Module 8: Advanced Regression Concepts

- Bias-Variance Trade-off
- Multiple Linear Regression
- Polynomial Regression
- Task 4

Module 9: Classification

- Logistic Regression
- Classification Metrics
- Imbalanced Dataset
- Softmax Regression
- Cross-Validation
- K-Fold Cross-Validation
- Project 1

Module 10: Linear Algebra and Statistics

- Linear Algebra
- Probability
- Statistics

Module 11: Linear Regression Assumptions

- Assumptions of Linear Regression

Module 12: Dimensionality Reduction

- Curse of Dimensionality
- PCA
- Feature Selection
- Task 5

Module 13: Classical Machine Learning Algorithms

- SVM
- KNN
- K-Means Clustering
- Clustering Evaluation
- Task 6

Module 14: Tree-Based and Ensemble Learning

- Decision Trees and Regression Trees
- Voting
- Bagging
- Random Forest
- Boosting
- XGBoost
- Stacking
- Task 7

Module 15: Regularization Techniques

- Regularization
- Ridge Regression
- Lasso Regression
- Elastic Net

Module 16: Introduction to Deep Learning

Module 17: Major Project and Evaluation for Internship

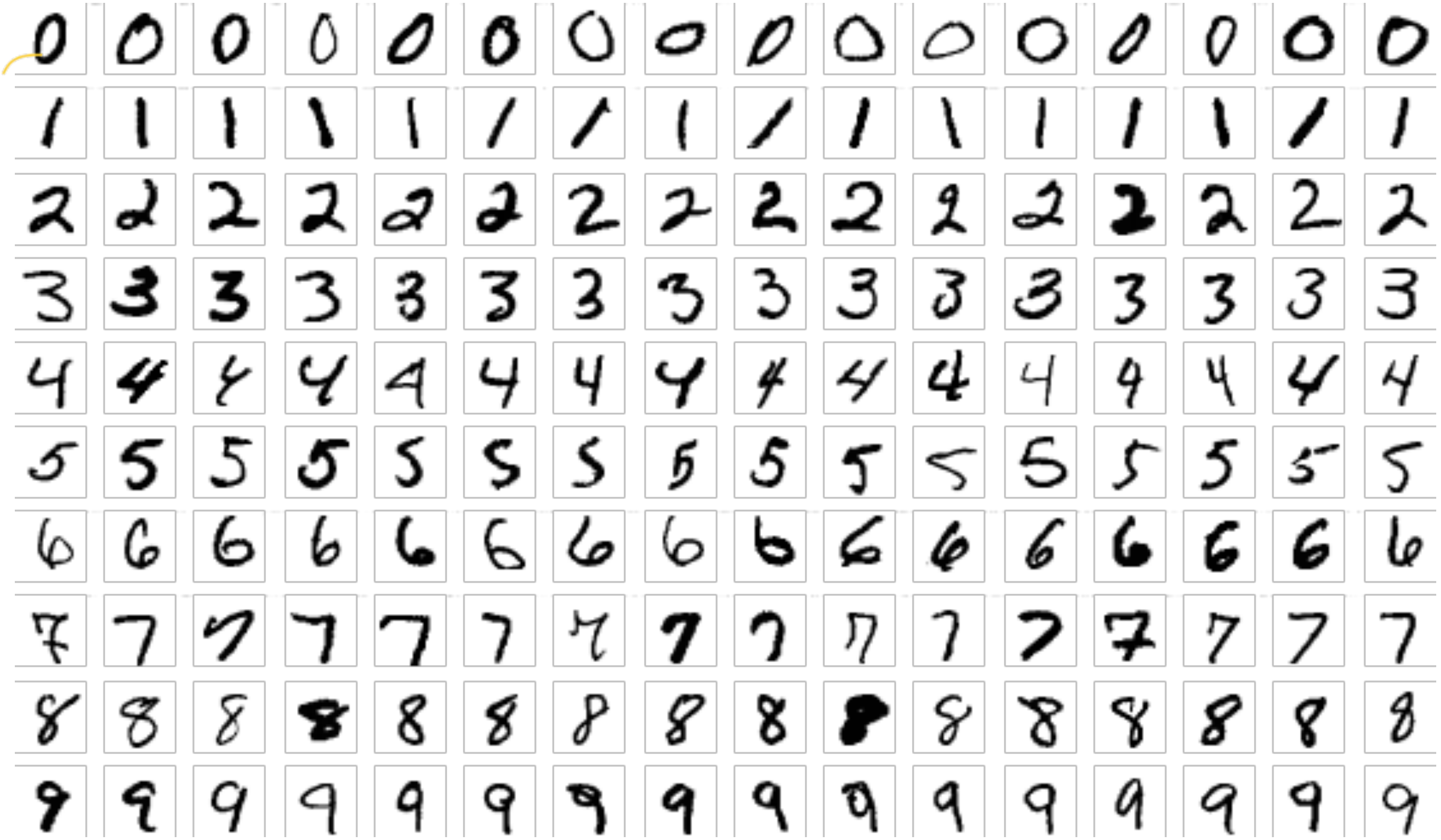
Why Machine Learning?

Why ML?

- Suppose, I want to identify different digits from hand-written text

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Why ML? An Example



Why ML? Another Example

How to differentiate?

Body colour?

https://www.business-standard.com/india-news/happy-international-zebra-day-2024-history-importance-quotes-and-wishes-124013100260_1.html

https://en.wikipedia.org/wiki/Tiger

Why ML? Another Example

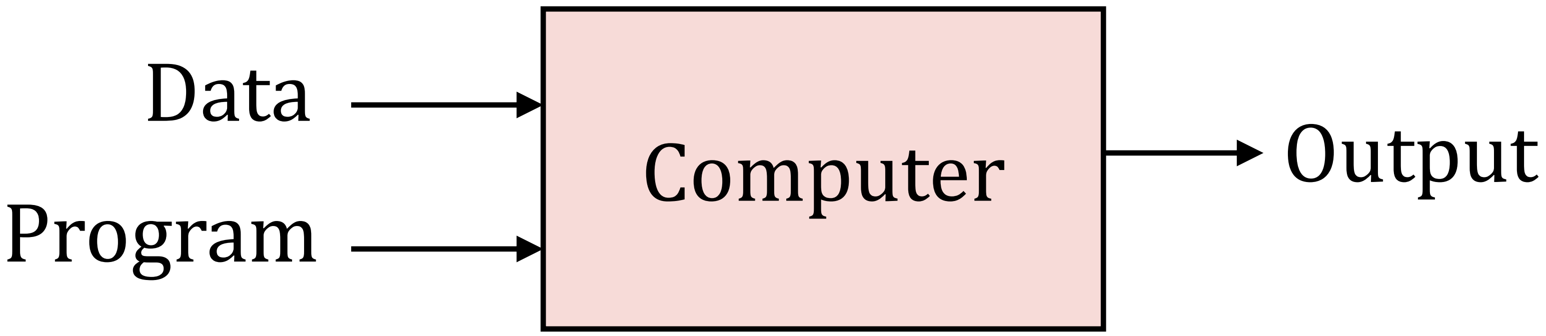
How to differentiate?

Body colour? Then, what is this animal?

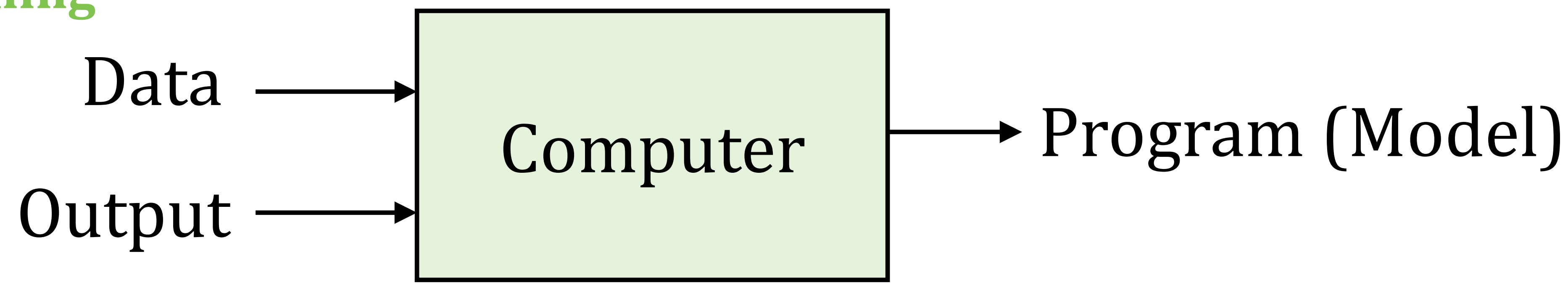
https://en.wikipedia.org/wiki/White_tiger

What is Machine Learning?

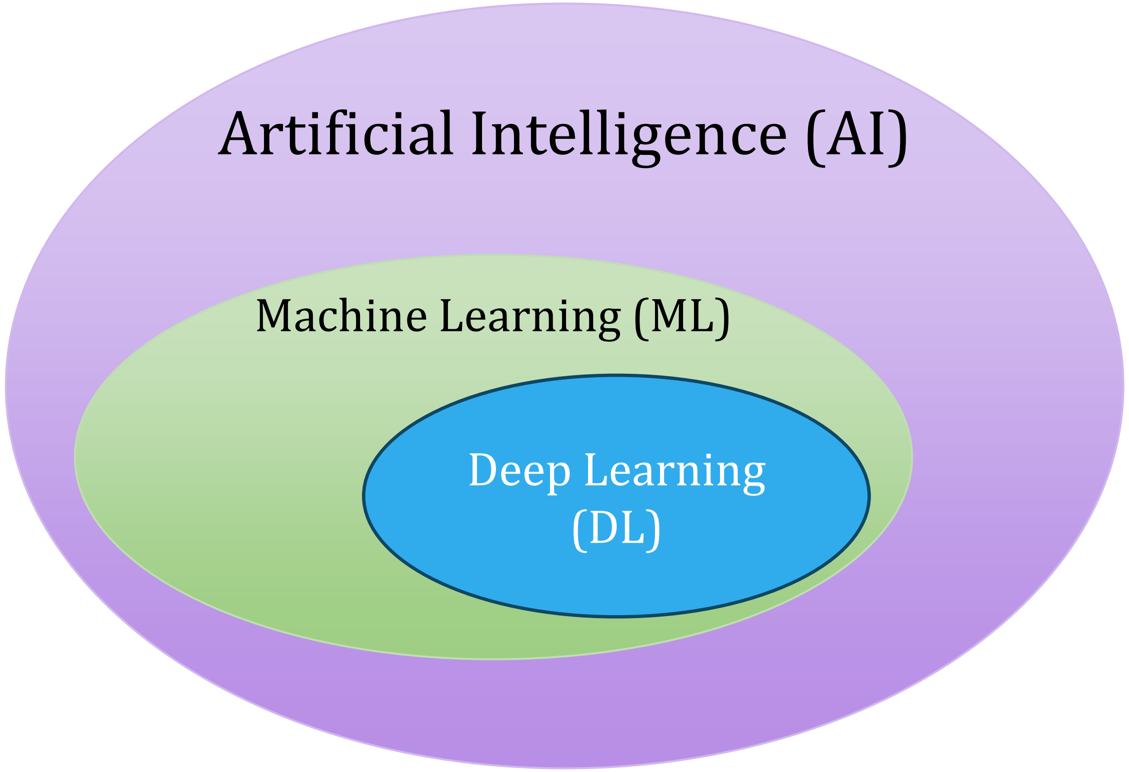
Traditional Programming



Machine Learning



What is Machine Learning



History of ML

https://medium.com/@marizombie/important-dates-in-history-of-machine-learning-d52d7f24b797

ML Today

What’s New Today?

- Availability of hardware
  - GPUs
  - Storage
  - Cloud services

- Availability of large datasets

- Availability of implementation platforms
  - PyTorch
  - Tensorflow

Different Types of ML

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

* Supervised Learning :

↳ Data + label

Supervised Learning

- Mainly of two types
  - Classification
    - Labels are categorical in nature
      - (Cat, dog, elephant), (foreground, background), etc.
  - Regression
    - Labels are real numbers
      - Price of house, Hemoglobin (in gm%), coordinate of a robot, etc.

Different Types of ML

- Supervised Learning
- **Unsupervised Learning**
- Reinforcement Learning

Unsupervised Learning

https://www.pexels.com/

Unsupervised Learning

Unsupervised Learning




Only data and no labels



Unsupervised Learning

- Making sense of data
- Input to learner (input for experience)
  - Data
  - No label
- Output: Target task such as clustering

Unsupervised Learning Example

- Clustering
  - Making cohesive group of data points
  - Example application
    - Recommendation system based on customer’s pervious choices
    - Finding similar viruses based on genetic pattern
- Dimensionality reduction
  - Application: finding salient features in data
  - Data compression
- Association rule mining
  - Finding cooccurrence
- Generative Models
- Unsupervised feature extraction

Different Types of ML

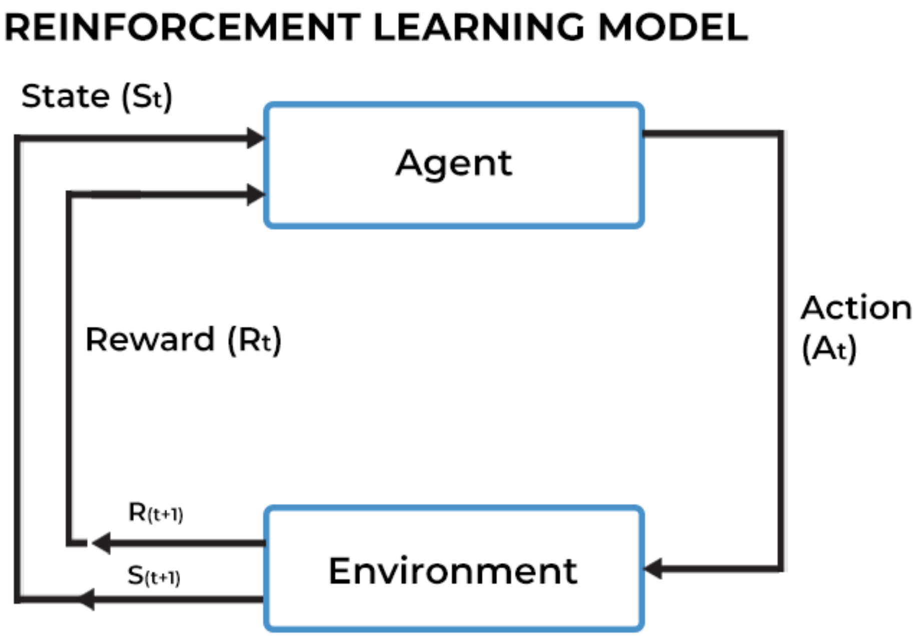
- Supervised Learning
- Unsupervised Learning
- **Reinforcement Learning**

Reinforcement Learning

How did you learn to ride bicycle?

https://ptscoaching.com/2015/04/teaching-your-child-life-skills/

Reinforcement Learning



What do we learn?
To control an agent (bicycle) through **a sequence of** good decisions (actions)

How do we learn it?
From experiences through rewards and penalty from the environment

https://www.spiceworks.com/tech/artificial-intelligence/articles/what-is-reinforcement-learning/